IILM UNIVERSITY
Greater Noida, Uttar Pradesh

INTERNSHIP EVALUATION PRESENTATION

Customer Shopping Behavior Analysis using Python, SQL & Power BI

AICTE – EduSkills Virtual Internship | Data Analytics with Python & Power BI

STUDENT NAME
Ali Ghazanfar

ROLL NUMBER
CS-2341887

PROGRAM & BATCH
B.Tech CSE — Batch 2023-27

ORGANIZATION
AICTE – EduSkills Academy

INTERNSHIP DURATION
August 2026 – October 2026

INSTITUTION
IILM University, Greater Noida

AGENDA

What this presentation covers

An 8-week AICTE–EduSkills virtual internship in Data Analytics with Python & Power BI, culminating in a capstone project.

01 Organization Profile — AICTE, National Internship Portal & EduSkills

02 Internship Overview & Objectives

03 Scope, Technologies & System Architecture

04 Week-wise Learning Journey (Weeks 1–8)

05 Capstone Project — Customer Shopping Behavior Analysis

06 Key Results & Business Recommendations

07 Challenges Faced & Key Learnings

08 Outcomes, Certification & Conclusion

SECTION 01

Organization Profile

1

AICTE

All India Council for Technical Education, Ministry of Education, Government of India — the apex body regulating technical education in the country.

2

National Internship Portal

A Ministry of Education initiative connecting technical-institution students with structured, industry-recognized virtual internships.

3

EduSkills Academy

ISO 9001:2015 & ISO 20000-1:2018 certified ed-tech partner delivering the curriculum, mentorship, weekly assessments and the capstone project.

The internship was formally structured with a signed offer letter, an 8-week curriculum, weekly assessments, a capstone project, a Final Assessment Test, and a verifiable digital completion certificate.

SECTION 02

Internship Overview

Domain	Data Analytics with Python & Power BI
Duration	8 Weeks (August 2026 – October 2026)
Mode	Virtual, self-paced with weekly structured modules
Evaluation	Weekly assessments, project documentation, Final Assessment Test
Certification	Virtual Internship Certificate (AICTE – EduSkills), issued on completion
Stipend	Not applicable (learning-focused program)

SECTION 02

Project Objectives

1. Clean, prepare and explore a raw customer shopping transactional dataset using Python
2. Perform structured business-question analysis using SQL on a relational (PostgreSQL) database
3. Identify revenue patterns across gender, age group, subscription status and shipping type
4. Segment customers into New, Returning and Loyal categories based on purchase history
5. Identify top-performing and discount-dependent products
6. Design an interactive Power BI dashboard summarizing key business metrics
7. Translate analytical findings into actionable business recommendations

SECTION 03

Scope & Technologies

PROJECT SCOPE

A retail transactional dataset of 3,900 rows and 18 columns — covering demographics, purchase details and shopping behavior — taken through the full analytics pipeline:

- Data ingestion & cleaning in Python
- Feature engineering
- Database integration with PostgreSQL
- SQL-based business analysis
- Dashboard development in Power BI

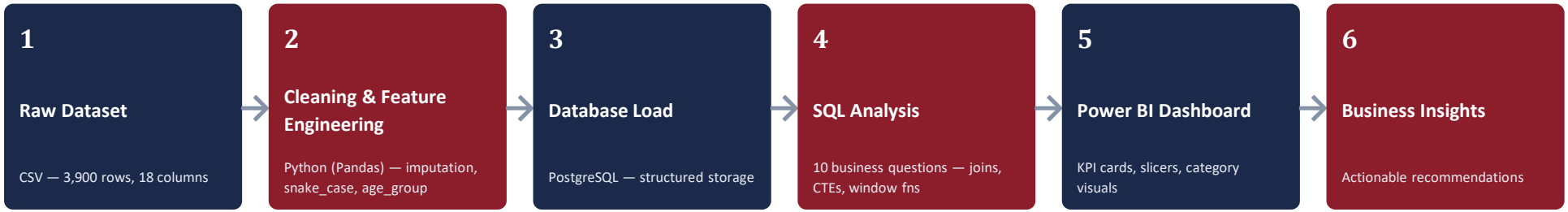
Limited to the given dataset; real-time pipelines and predictive modeling are noted as future extensions.

Programming Language	Python (Pandas, NumPy)
Data Visualization (EDA)	Matplotlib, Seaborn
Database	PostgreSQL
Query Language	SQL — subqueries, CTEs, window functions, joins
Business Intelligence	Microsoft Power BI
Dev Environment	Jupyter Notebook / Anaconda

SECTION 03

System Architecture & Workflow

A linear, full-stack analytics workflow — raw data flows through cleaning, storage, querying and visualization to reach business insight.



METHODOLOGY HIGHLIGHTS

- 37 missing Review Rating values imputed using the median rating per product category
- Redundant promo_code_used column identified and dropped after a data-consistency check
- Feature engineering added age_group and purchase_frequency_days for segmentation

SECTION 04

Week-wise Learning Journey — Weeks 1–4

W1	**Data Analytics Foundations & Python Basics**	Course & environment setup (Python/Anaconda, SQL, Power BI); core Python — variables, control flow, functions, NumPy and Pandas fundamentals.
W2	**Data Cleaning & SQL Fundamentals**	Data preprocessing with Pandas (missing values, outliers, transformations); SQL basics including SELECT, WHERE, GROUP BY, and joins.
W3	**Advanced SQL & Power BI Basics**	Subqueries, CTEs, window functions, DML operations; introduction to the Power BI interface, data connections and basic visualizations.
W4	**Python–SQL Integration & Data Modeling**	Python database connections and executing queries, automation workflows; Power BI data modeling, DAX basics and time-intelligence functions.

SECTION 04

Week-wise Learning Journey — Weeks 5–8

W5	**Advanced Visualization with Python & Power BI**	Advanced Power BI visuals (drill-through, bookmarks, filters), performance optimization; Python visualization using Matplotlib and Seaborn.
W6	**Integration, Publishing & Collaboration**	Python-scripting integration in Power BI, advanced use cases; publishing dashboards to Power BI Service, sharing and workspace management.
W7	**Real-World Projects & Workflow Best Practices**	Sales performance analysis, customer behavior segmentation, structured workflow practices, and best practices in Python, SQL and Power BI.
W8	**Capstone Project — Analysis, Dashboard & Delivery**	E-commerce dataset cleaning, SQL-based extraction and transformation, KPI identification, dashboard building, insight generation and final presentation.

SECTION 05 · CAPSTONE PROJECT

Customer Shopping Behavior Analysis

PROBLEM STATEMENT

Retail businesses generate large volumes of transactional data that rarely translate into actionable insight without structured analysis. The project builds a clear picture of who the customers are, how they shop, which products perform best, and how discounts and subscriptions affect revenue — to guide targeted marketing, inventory and loyalty strategies.

METHODOLOGY

- Loaded & explored 3,900 rows × 18 columns with Pandas
- Imputed 37 missing ratings via category median; standardized columns
- Engineered age_group and purchase_frequency_days features
- Loaded cleaned data into PostgreSQL for SQL-based analysis
- Answered 10 business questions in SQL; visualized in Power BI

3,900
Customer Purchases

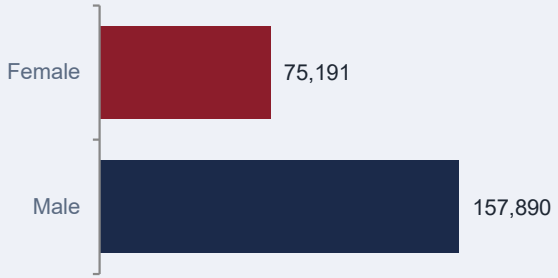
18
Data Columns

10
SQL Business Questions

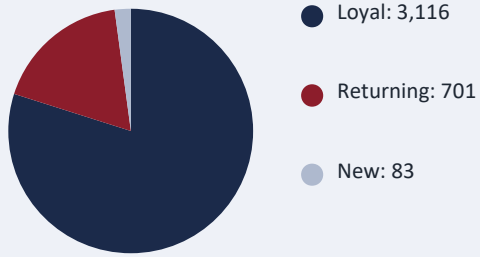
SECTION 05 · CAPSTONE PROJECT

Key Results

Revenue by Gender (₹)



Customer Segmentation



Subscription Impact

₹1,70,436

from 2,847 non-subscribers

₹62,645

from 1,053 subscribers

Average spend per customer is similar across both groups

839

high-spending discount users

47–50%

discount share on Hat, Sneakers, Coat, Sweater, Pants

₹60.48 vs ₹58.46

avg. Express vs Standard shipping spend

Young Adult

top revenue age group (₹62,143)

SECTION 05 · CAPSTONE PROJECT

Business Recommendations

1

Boost Subscriptions

Drive subscription uptake through exclusive member benefits.

2

Loyalty Programs

Introduce programs that convert Returning customers into Loyal ones.

3

Review Discount Policy

Balance sales growth against profit margins on discount-heavy products.

4

Feature Top Products

Prominently market top-rated and best-selling products.

5

Targeted Marketing

Focus campaigns on high-revenue age groups and express-shipping users.

SECTION 06

Challenges Faced & Key Learnings

CHALLENGES

- Deciding a fair way to impute 37 missing review ratings without skewing category averages
- Spotting and removing a redundant column (promo_code_used) that duplicated existing information
- Keeping the Python → PostgreSQL → Power BI pipeline consistent across every stage
- Designing a segmentation logic (New / Returning / Loyal) that stayed meaningful for the business
- Translating raw SQL outputs into a clean, filterable Power BI dashboard

KEY LEARNINGS

- Clean, well-documented feature engineering pays off later in SQL and Power BI
- Median-based imputation is a simple, defensible way to handle small pockets of missing data
- Structuring business questions before writing SQL keeps queries focused and reusable
- A good dashboard is built around the decisions it needs to support, not just the data available
- Data analytics is as much about disciplined workflow as it is about syntax

SECTION 07

Outcomes, Certification & Conclusion

SKILLS GAINED

- Python for Data Analysis (Pandas, NumPy)
- SQL & Relational Databases (PostgreSQL)
- Data Cleaning & Feature Engineering
- Business Intelligence Dashboarding (Power BI)
- Data-Driven Business Recommendations

CERTIFICATION STATUS

OFFER LETTER REF. NO.

C18Y2026M081546344

AICTE STUDENT ID

STU6997660ac608d1771529738

STRUCTURE

Weekly assessments + project docs + Final Assessment Test

STATUS

Certificate issued on successful completion of all assessments

This internship built hands-on, end-to-end analytics capability — from raw data to a deployable Power BI dashboard — while reinforcing SQL, database design and business-communication skills for real-world decision-making.